

The Physics of the Schwarzschild Radius: Foundations of Relativistic Astrophysics

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1 Historical Background

The formula of the Schwarzschild radius was created by the outstanding German astronomer and physicist Karl Schwarzschild, who laid the foundations of the mathematical apparatus of modern relativistic astrophysics.

In 1915, just a month after Albert Einstein published his General Theory of Relativity, Schwarzschild found the first exact analytical solution of the Einstein field equations for a static, spherically symmetric mass distribution (known today as the *Schwarzschild metric*). This groundbreaking solution theoretically predicted the existence of event horizons and the critical radius of gravitational collapse, which became the absolute foundation of black hole physics.

2 The Mathematical Formulation

Today we will analyze the mathematical structure of this phenomenon. The definitive formula for calculating the Schwarzschild radius is defined as follows:

$$R_s = \frac{2GM}{c^2} \quad (1)$$

Where the physical variables are defined as:

- R_s — The Schwarzschild radius (measured in meters, m);
- G — The Universal Gravitational constant ($\approx 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$);
- M — The total mass of the astronomical object (measured in kilograms, kg);
- c — The constant speed of light in a vacuum (299792458 m/s).

2.1 Real-World Examples

To better understand how extreme this gravitational compression is, we can look at what the Schwarzschild radius would be for well-known celestial bodies if they were to become black holes:

- **The Earth:** If we compressed the entire mass of our planet ($M_{\oplus} \approx 5.97 \times 10^{24}$ kg) into a black hole, its Schwarzschild radius would be approximately 8.87 mm (roughly the size of a marble or a small grape).
- **The Sun:** If our Sun ($M_{\odot} \approx 1.989 \times 10^{30}$ kg) collapsed into a singularity, its radius would be about 2.95 km (meaning the entire Sun would fit inside a small city).
- **A Human:** For an average person weighing 70 kg, the Schwarzschild radius would be roughly 1.04×10^{-25} meters — an scale trillions of times smaller than a single proton.

3 The Physical Essence and Significance

The core essence of the formula reveals to us how to calculate the exact radius of a black hole's event horizon. It dictates how much it is theoretically necessary to compress any physical object in order for its escape velocity to equal the speed of light, effectively turning it into a black hole.

This phenomenon practically occurs in the universe when a massive star runs out of its nuclear fuel. Once the internal pressure drops, the star is relentlessly compressed by its own intense gravity into the bounds of the Schwarzschild radius, resulting in the birth of a stellar-mass black hole.

4 Modern Legacy and Conclusion

The topic of black holes is incredibly large, deep, and still not fully disclosed by modern science. However, the Schwarzschild radius remains an integral, non-negotiable part of studying these spacetime singularities and their behavior.

Today, this foundational concept is heavily utilized in advanced theoretical physics, serving as a stepping stone into deeper topics such as:

1. The Schwarzschild metric and external spacetime curvature;
2. Quantum gravity and Hawking radiation;
3. Planck black holes (microscopic primordial black holes);
4. The Kerr metric (which describes rotating black holes).